

Gamma detection for source-localisation support in targeted alpha therapy

C. De Sio^a, L. Abu Sabah^a, J. Duan^a, Y. Shi^a, S. Williams^a, J. Velthuis^a

^aSchool of Physics, University of Bristol, Bristol, United Kingdom

Targeted alpha therapies can deliver high localised dose, but treatment planning and delivery depend on where the radionuclide source is located after administration. In the AlphaGlue delivery concept [1], α -emitters are applied to a tumor suspended in a thin glue layer. The activity of the loaded glue is known at the moment of production, but it is not trivial to measure it at the moment of application without affecting the treatment effectiveness. Here we present a simulation study using the decay-chain gamma detection that shows that it is possible to monitor the applied dose and the spread of the α -emitters of the decay chain through the body in clinically appropriate times. This means that gamma detection could support emitter-position monitoring, treatment planning, and delivery verification.

Introduction

In targeted alpha therapy, the delivered dose is strongly local. A treatment plan therefore depends not only on the initial activity distribution, but also on whether source material and progeny remain near the intended deposition site, diffuse into surrounding tissue, or are redistributed by biological transport[2, 3]. The AlphaGlue concept proposes suspending alpha emitters in a thin glue layer applied to tissue [1]. The concept has previously been evaluated with Geant4 and Geant4-DNA simulations for dose and DNA-damage modelling [4, 6]. This study investigates whether the accompanying gamma signal can be used as a theranostic observable, to monitor the emitter position, detect source movement, and provide useful information as a function of detection time.

Simulation model

Photon transport was simulated with the Geant4 toolkit [4]. The model places a 40 mm $\times$ 2 mm $\times$ 40 mm loaded glue layer on the $+y$ surface of a 20 cm $\times$ 20 cm $\times$ 150 cm water phantom. Six detector panels surround the phantom at 20 cm from the water surface. Each photon entry into a detector panel is counted as one hit.

Six source configurations were studied to separate three clinically relevant possibilities: emitter retained in the loaded layer, emitter broadly mixed into the surrounding volume, and emitter transported to a localised displaced region. The retained-source case is used as the baseline. As discussed in [2, 3], some of the α -emitters in the decay chain get taken up by the blood. [2, 3] report large variations in the amount of this in different experiments, further emphasizing the need for local dose verification. This is taken into account in the simulation by studying cases where 10%, 50%, or 90% of the radioisotope in the

water volume, while the localised-displacement cases place 20%, 40%, or 60% of the source in a small displaced water region. These configurations are intended as sensitivity tests for source movement and source localisation.

Each configuration was simulated with ten repetitions of 10,000 primary decays, which gives acceptable statistical uncertainties. These simulated decays were then rescaled to a DaRT-like reference activity of 3 μ Ci (111 kBq). For the simulated source half-life, this corresponds to approximately 5.3×10^{10} initial atoms, so each simulated primary decay represents approximately 5.3×10^6 real atoms. This activity is chosen to match the order of a single DaRT seed[5], which has a typical activity of 3 μ Ci. The simulated geometry and source regions are shown in Fig. 1.

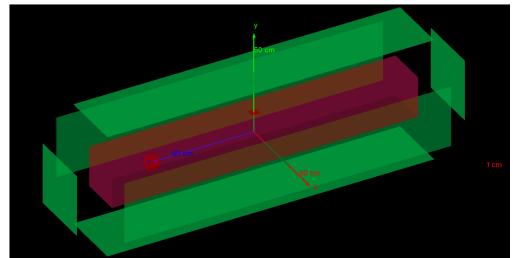


Figure 1: *Model geometry: water phantom (dark red), detector panels (green), and source regions (bright red). The loaded glue layer is placed on the $+y$ surface of the phantom; displaced-source scenarios move activity either into the water volume or to a localised region inside the phantom.*

Baseline subtraction

After rescaling to 3 μ Ci, the total expected detector yield changes only modestly between source configurations. In contrast, baseline-subtracted profiles show a clear spatial signature. Moving activity from the loaded layer into water reduces the dominant positive- y response and increases negative- y , lateral, and longitudinal responses. The residual profiles are candidate-minus-baseline detector-hit histograms after activity normalisation. The central applicator-facing signal decreased with source movement, falling to 0.94, 0.72, and 0.47 of baseline for the 10%, 50%, and 90% broad-redistribution cases, and to 0.84, 0.67, and 0.50 for the 20%, 40%, and 60% localised-displacement cases. This indicates that source movement is better detected from baseline-subtracted spatial hit patterns than from the total hit rate alone. The resulting y -panel profiles are shown in Fig. 2.

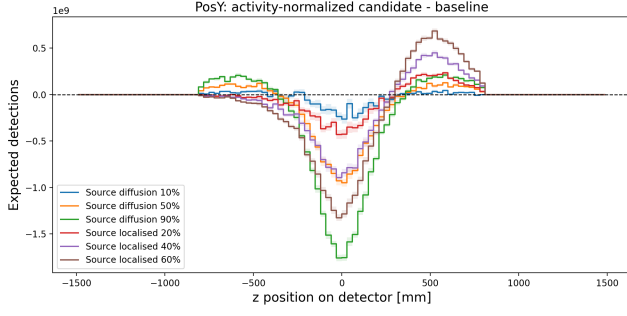


Figure 2: Activity-normalised candidate-minus-baseline z -profiles for the positive- y detector panel, scaled to a $3 \mu\text{Ci}$ reference source. Negative residuals indicate that moving activity away from the loaded layer reduces the expected hit yield in this panel relative to the retained-source baseline.

Signal inference

Using profile-shape features from repeated simulations, the broad redistribution fraction was recovered with a leave-one-out mean absolute error of 1.2 percentage points ($R^2 = 0.998$), as shown in Fig. 3. A separate secondary-source metric identified the localised displaced source in all 20%, 40%, and 60% localised-displacement cases. The reconstructed displaced-source position was centred at approximately $z = 525\text{--}531$ mm, compared with the simulated $z = 500$ mm source region. The corresponding source-position summary is shown in Fig. 4.

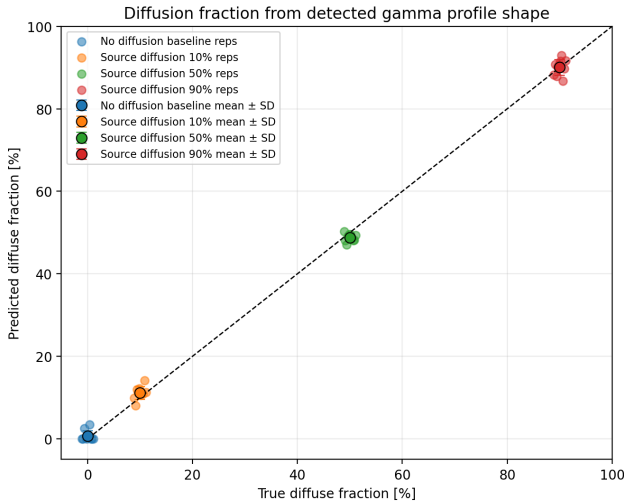


Figure 3: Cross-validated inference of the broad redistribution fraction from detector-hit profile features. The predicted fraction tracks the simulated 0%, 10%, 50%, and 90% redistribution cases with a leave-one-out mean absolute error of 1.2 percentage points.

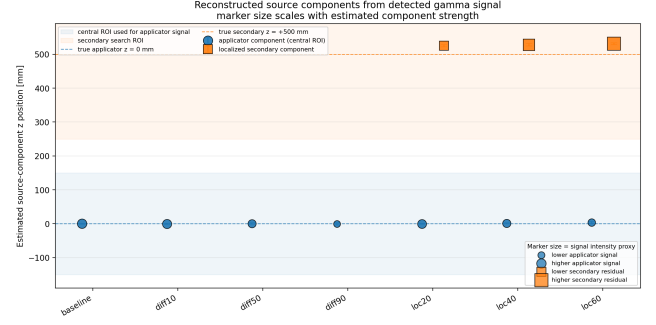


Figure 4: Inferred source-position summary from activity-normalised detector-hit residuals. The retained applicator component remains near the loaded layer, while localised-displacement cases produce a secondary component near the simulated displaced-source region.

Time dependence

Cumulative time-window profiles show that the baseline-subtracted signal is already visible on hour-scale windows and becomes stronger over longer observation windows. The detection time depends on the size and type of source movement, as summarized in Table 1. At the $3 \mu\text{Ci}$ activity scale, several redistribution observables crossed a 3σ threshold within 30 minutes to 2 hours for the larger source-movement cases. The weaker 10% broad redistribution case required about 6 hours, while the 20% localised displaced source produced a detectable displaced-region excess within about 1 hour. A $t < 10^6$ s analysis cut suppresses a late decay-chain tail while retaining the early signal relevant to the simulated source-localisation window.

Table 1: Representative first detection times after activity normalisation to a $3 \mu\text{Ci}$ source. Times are the first cumulative window at which the listed detector-hit observable crossed the given significance threshold.

Case	Metric	3σ	5σ
10% broad	y balance	6 h	6 h
50% broad	y balance	2 h	2 h
90% broad	y balance	30 min	1 h
20% localised	displaced excess	1 h	2 h
40% localised	displaced excess	30 min	1 h
60% localised	displaced excess	30 min	1 h

Outlook

These results indicate that decay-chain gamma detection can provide source-location information that is not captured by total count rate alone. Further work will include detector response, backgrounds, and patient-scale geometries so that the activity-normalised profiles can be interpreted as experimentally measurable count rates.

References

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